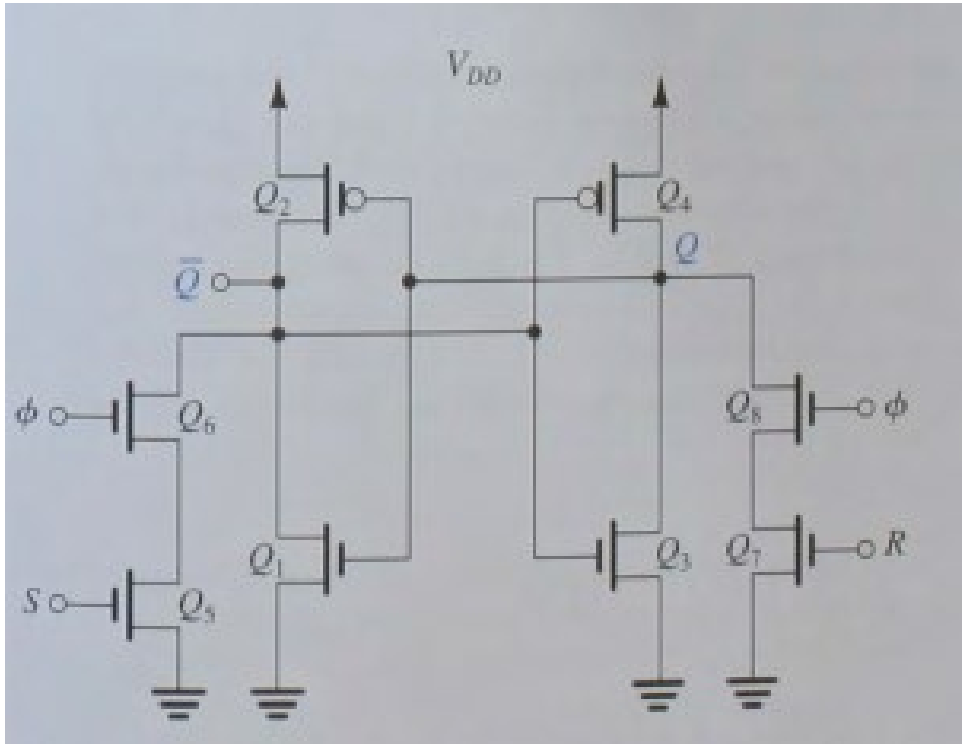
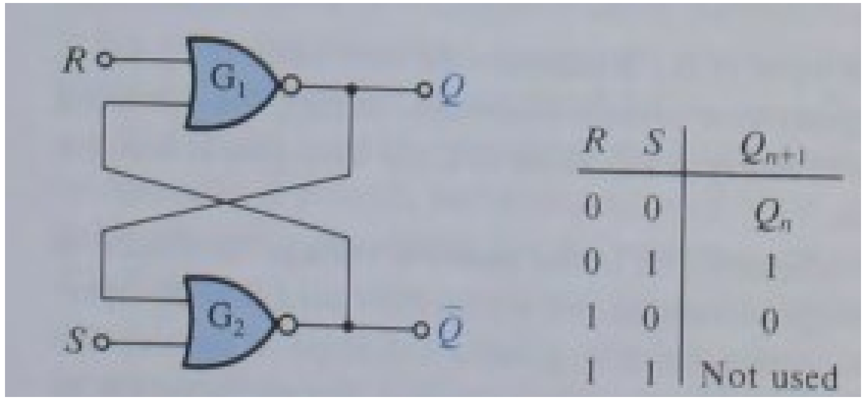
Applications of CMOS: Flip-Flops



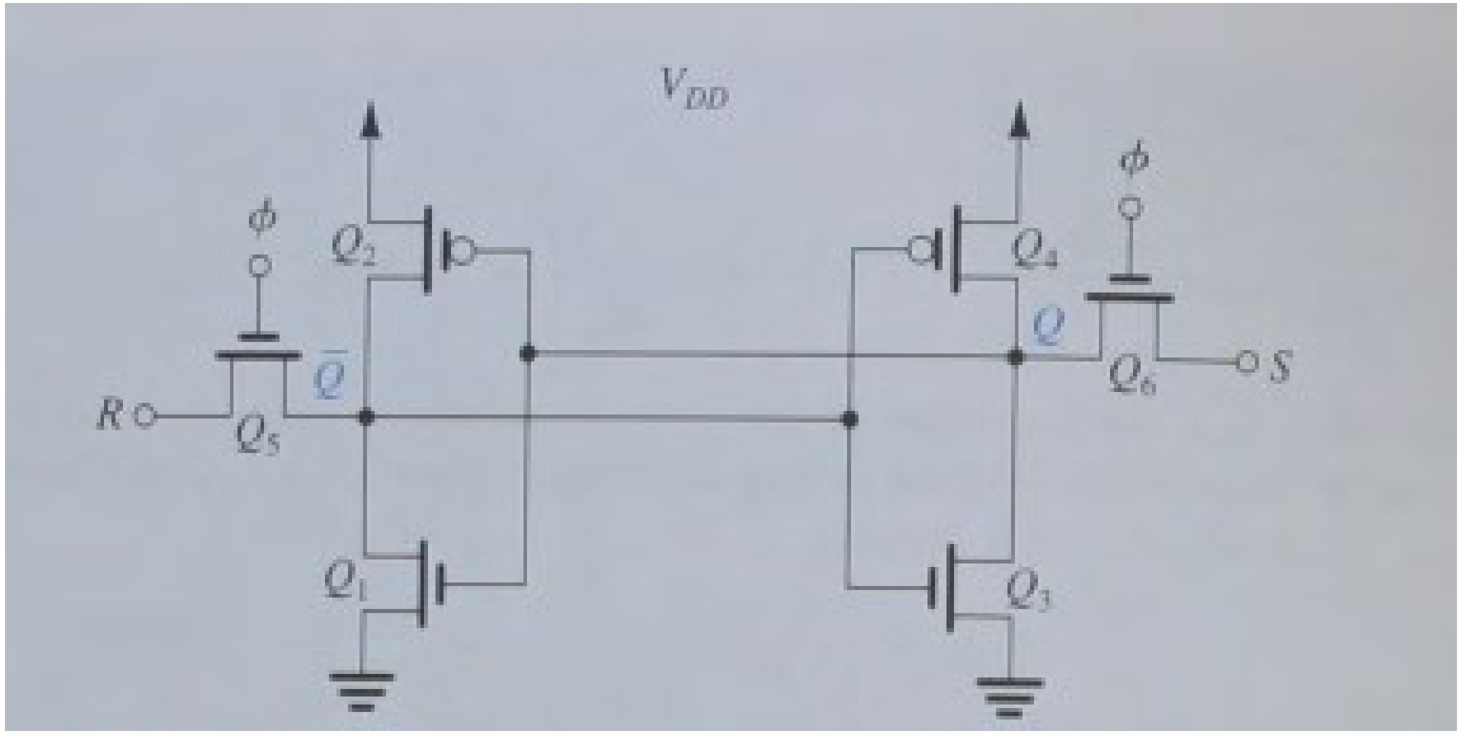
Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.



Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.

φ: clock signal
S: set signal
R: reset signal

Applications of CMOS: Flip-Flops as Memories



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

SRAM cell

φ: clock signal
S: set signal
R: reset signal

DRAM cell:
Gatekeeper transistor leading to capacitor storing the cell's value as charge

Applications of CMOS: CMOS Amplifiers

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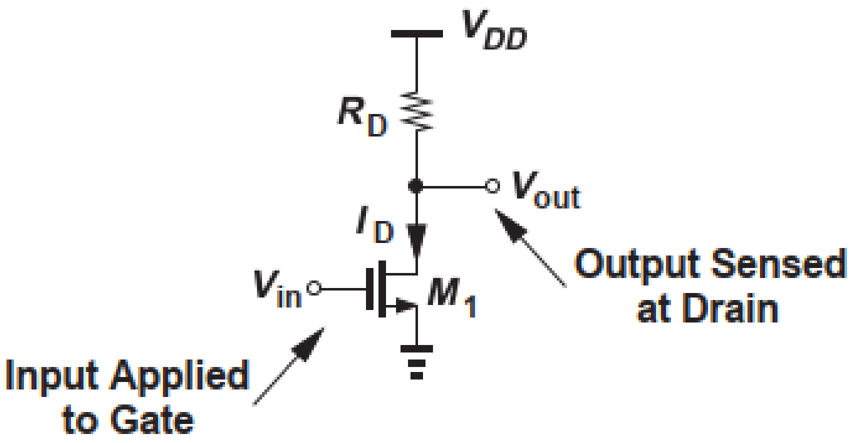
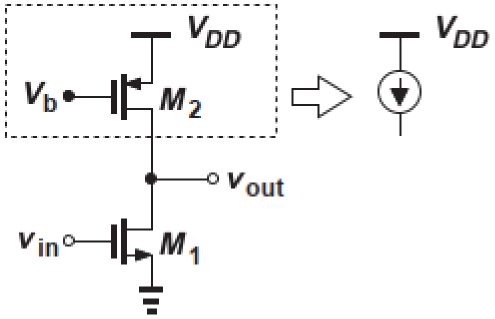
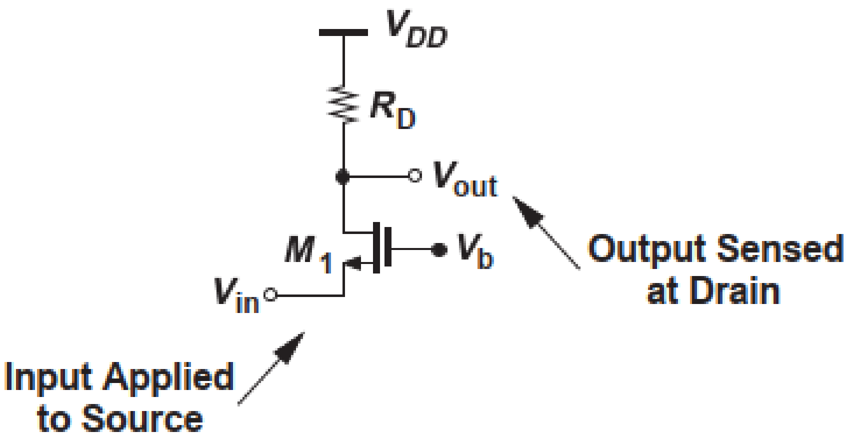
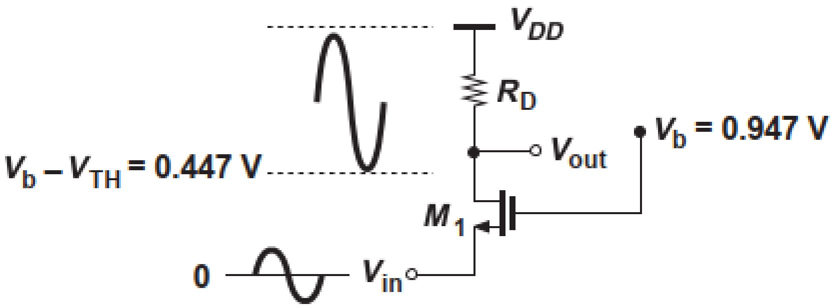




- Common Source

Razavi: “Fundamentals of Microelectronics”, 3rd edition, Wiley & Sons, 2021.

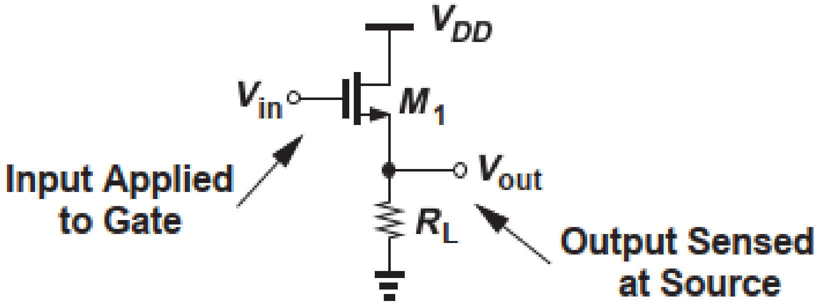




- Common Gate

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

Applications of CMOS: CMOS Amplifiers



Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

- Common Drain or Source Follower

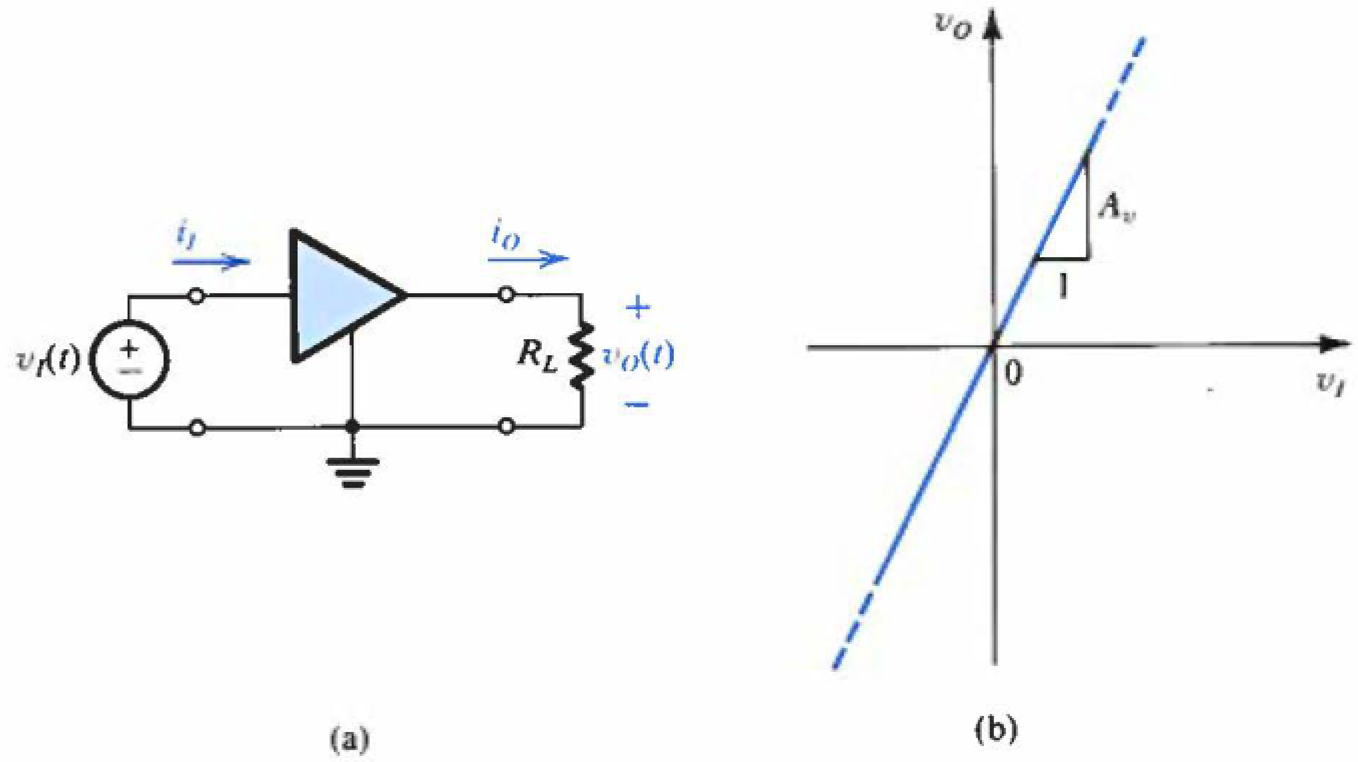


Figure 1.12 (a) A voltage amplifier fed with a signal $v_I(t)$ and connected to a load resistance R_L. (b) Transfer characteristic of a linear voltage amplifier with voltage gain A_v.

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Voltage Amplifiers

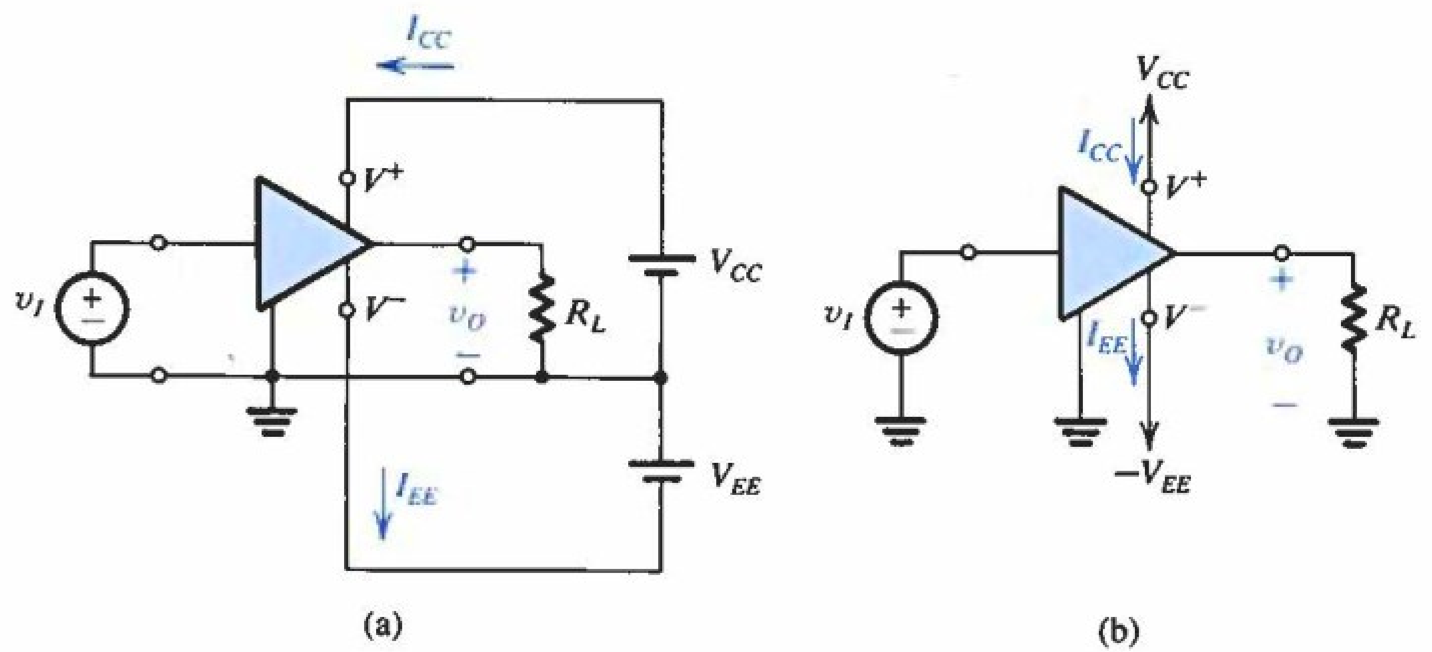
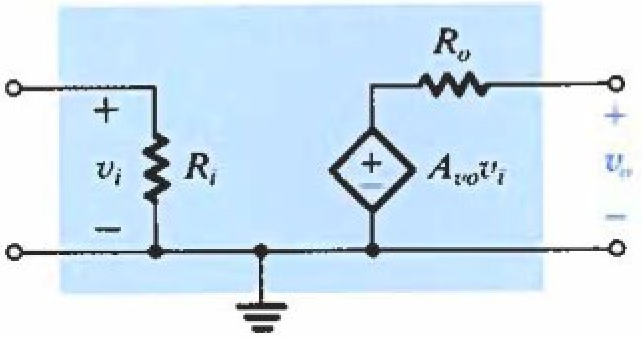


Figure 1.13 An amplifier that requires two dc supplies (shown as batteries) for operation.

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



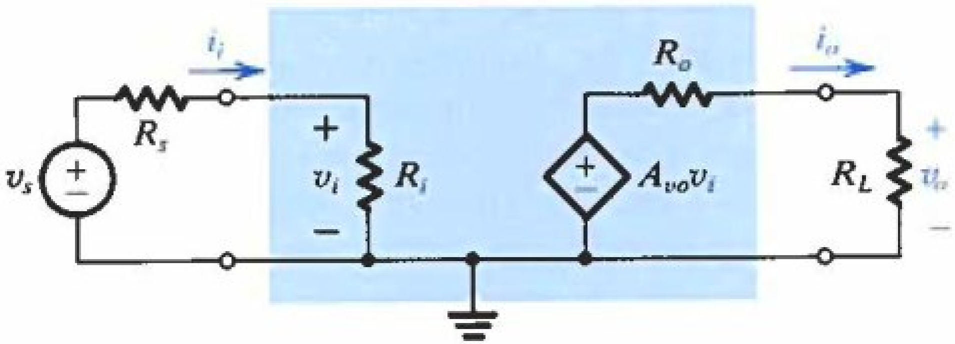
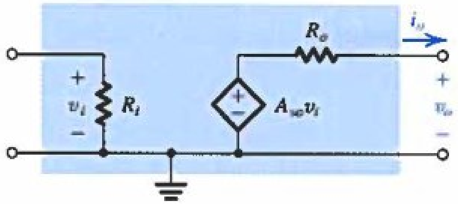
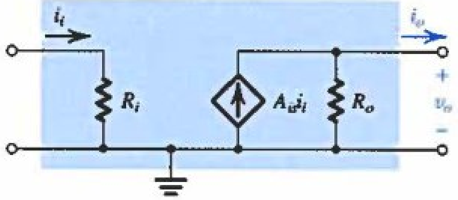
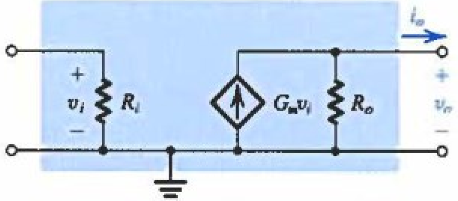
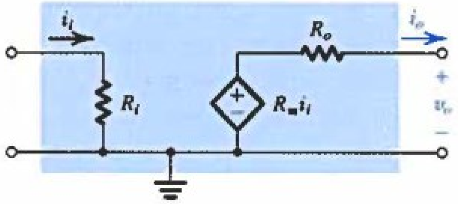


Figure 1.16 (a) Circuit model for the voltage amplifier. (b) The voltage amplifier with input signal source and load.

Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.

Amplifier Circuits

Table 1.1 The Four Amplifier Types

Type	Circuit Model	Gain Parameter	Ideal Characteristics
Voltage Amplifier		Open-Circuit Voltage Gain $A_{vo} \equiv \left.\frac{v_o}{v_i}\right\vert_{i_o=0}$ (V/V)	$R_i = \infty$ $R_o = 0$
Current Amplifier		Short-Circuit Current Gain $A_{is} \equiv \left.\frac{i_o}{i_i}\right\vert_{v_o=0}$ (A/A)	$R_i = 0$ $R_o = \infty$
Transconductance Amplifier		Short-Circuit Transconductance $G_m \equiv \left.\frac{i_o}{v_i}\right\vert_{v_o=0}$ (A/V)	$R_i = \infty$ $R_o = \infty$
Transresistance Amplifier		Open-Circuit Transresistance $R_m \equiv \left.\frac{v_o}{i_i}\right\vert_{i_o=0}$ (V/A)	$R_i = 0$ $R_o = 0$

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Operational Amplifiers

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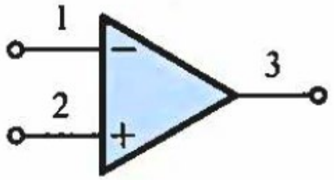


Figure 2.1 Circuit symbol for the op amp.

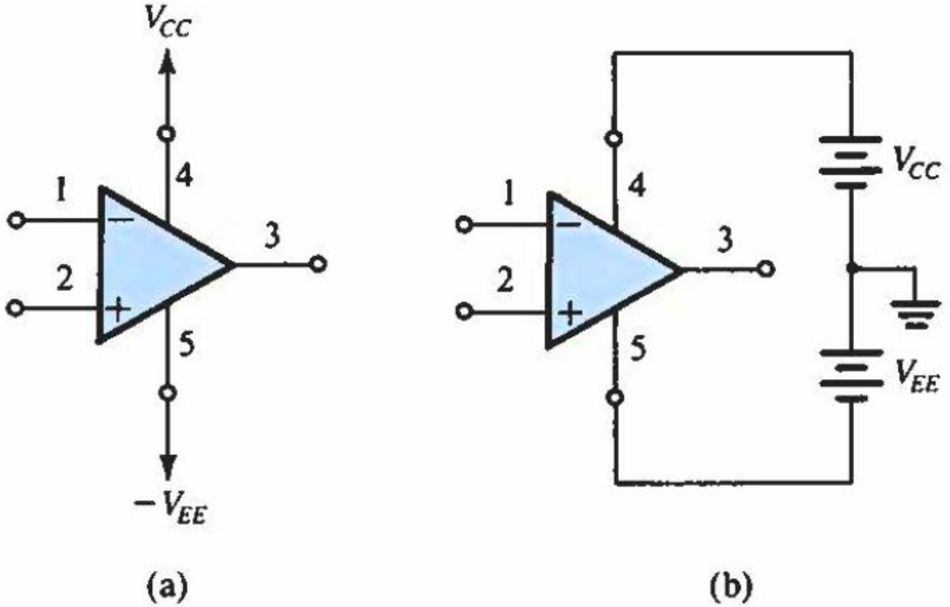
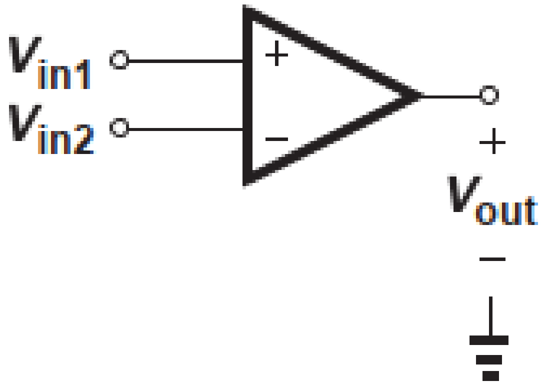


Figure 2.2 The op amp shown connected to dc power supplies.

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

- Operational amplifiers allow for different (versatile) applications
- Real-world integrated circuits perform quite closely to the ideal ones



$$V_{out} = A_0(V_{in1} - V_{in2})$$

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

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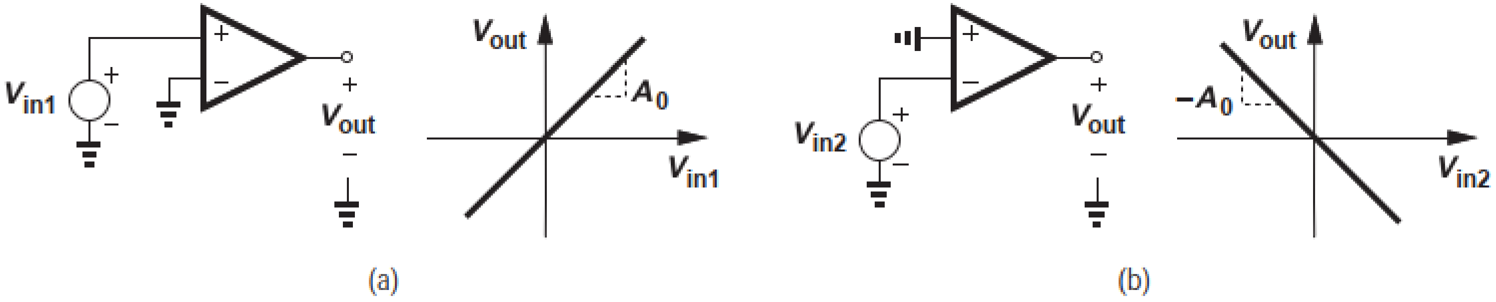
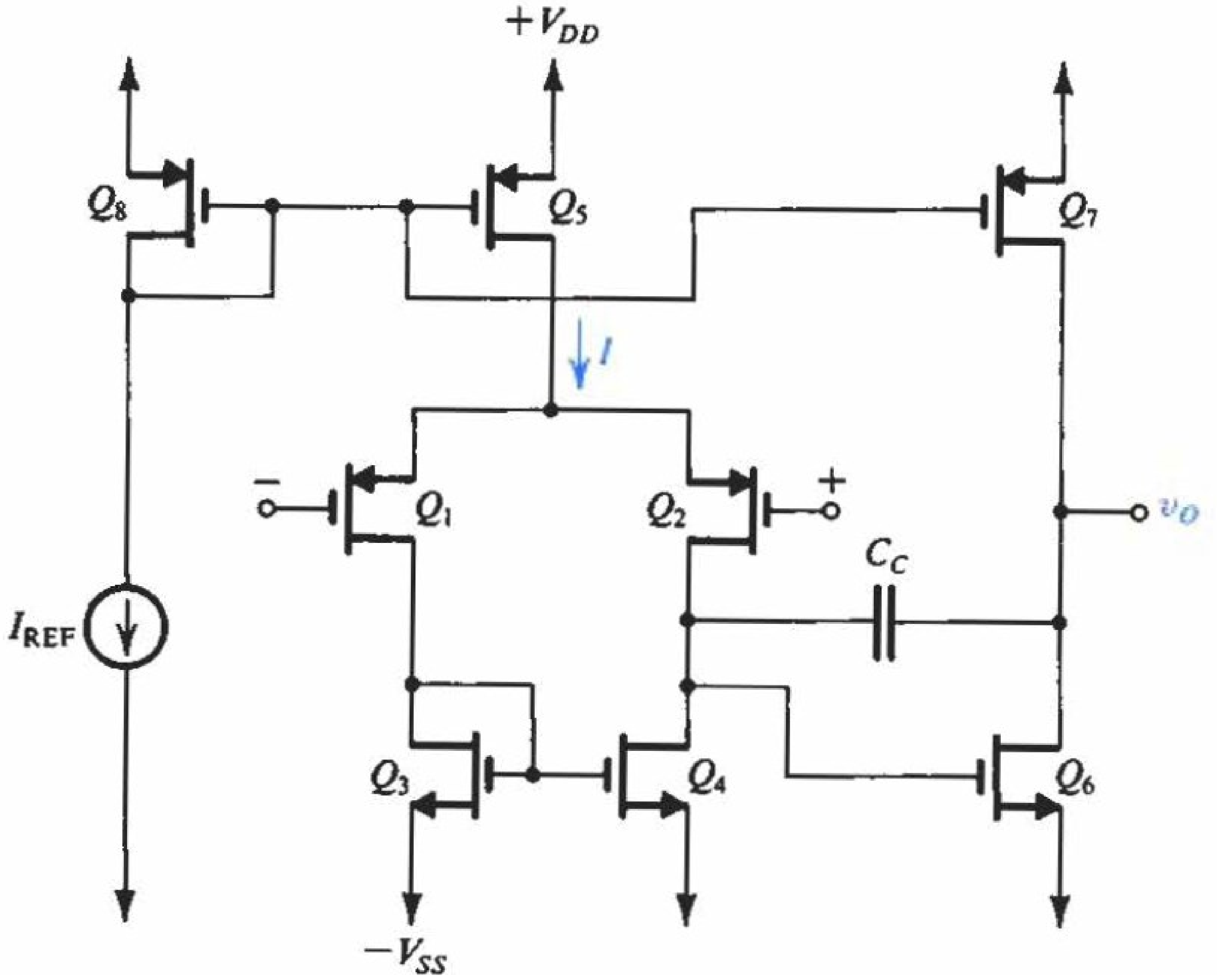


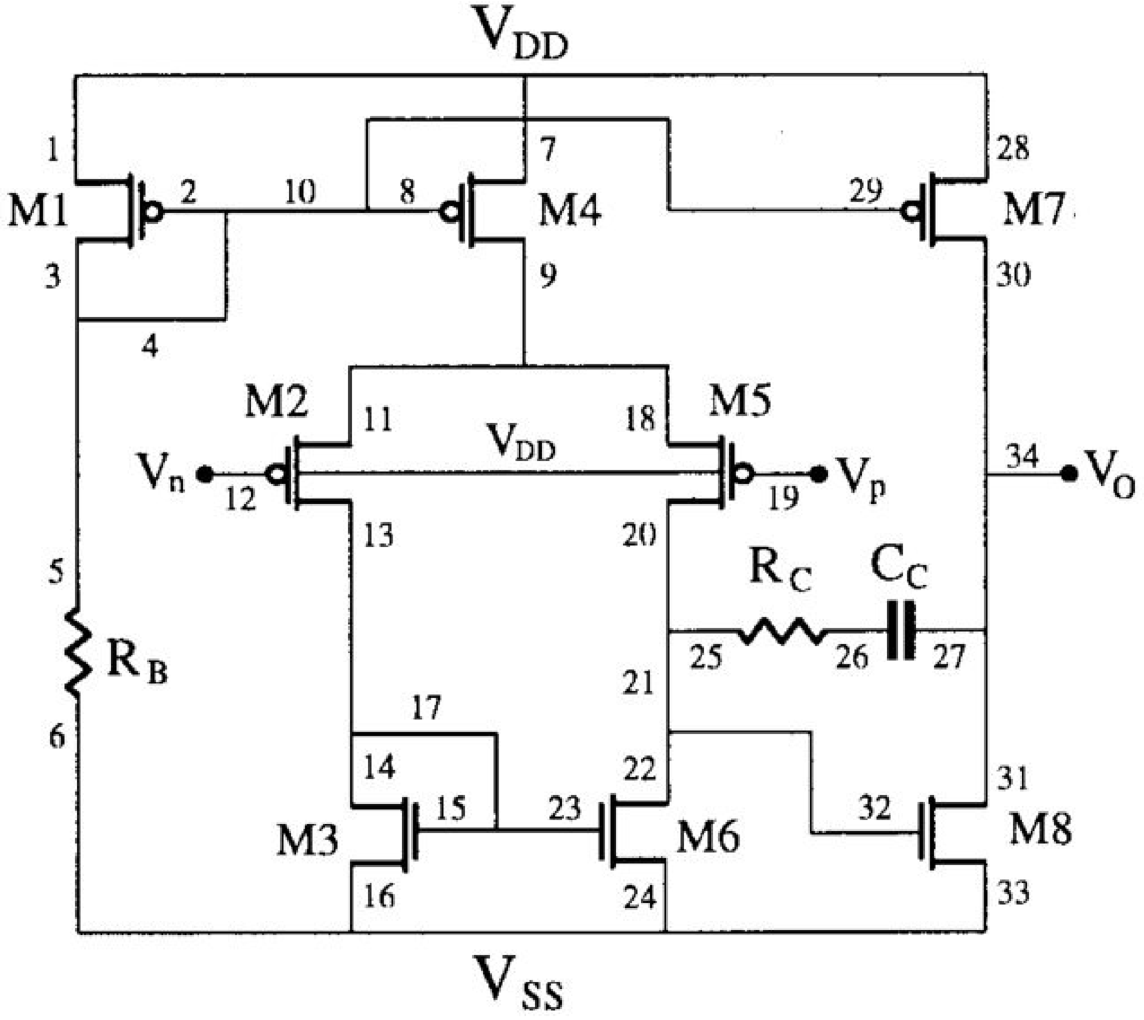
Figure 8.2 Op amp characteristics from (a) noninverting and (b) inverting inputs to output.

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.



Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.

- Need for a lot of (eight) transistors and one capacitor



- Need for a lot of (eight) transistors, one capacitor and two resistors
- Compensated CMOS operational amplifier

https://www.researchgate.net/profile/Bozena-Kaminska-2/publication/2977686/figure/fig4/AS:668711048458246@1536444546312/Compensated-CMOS-operational-amplifier_W640.jpg

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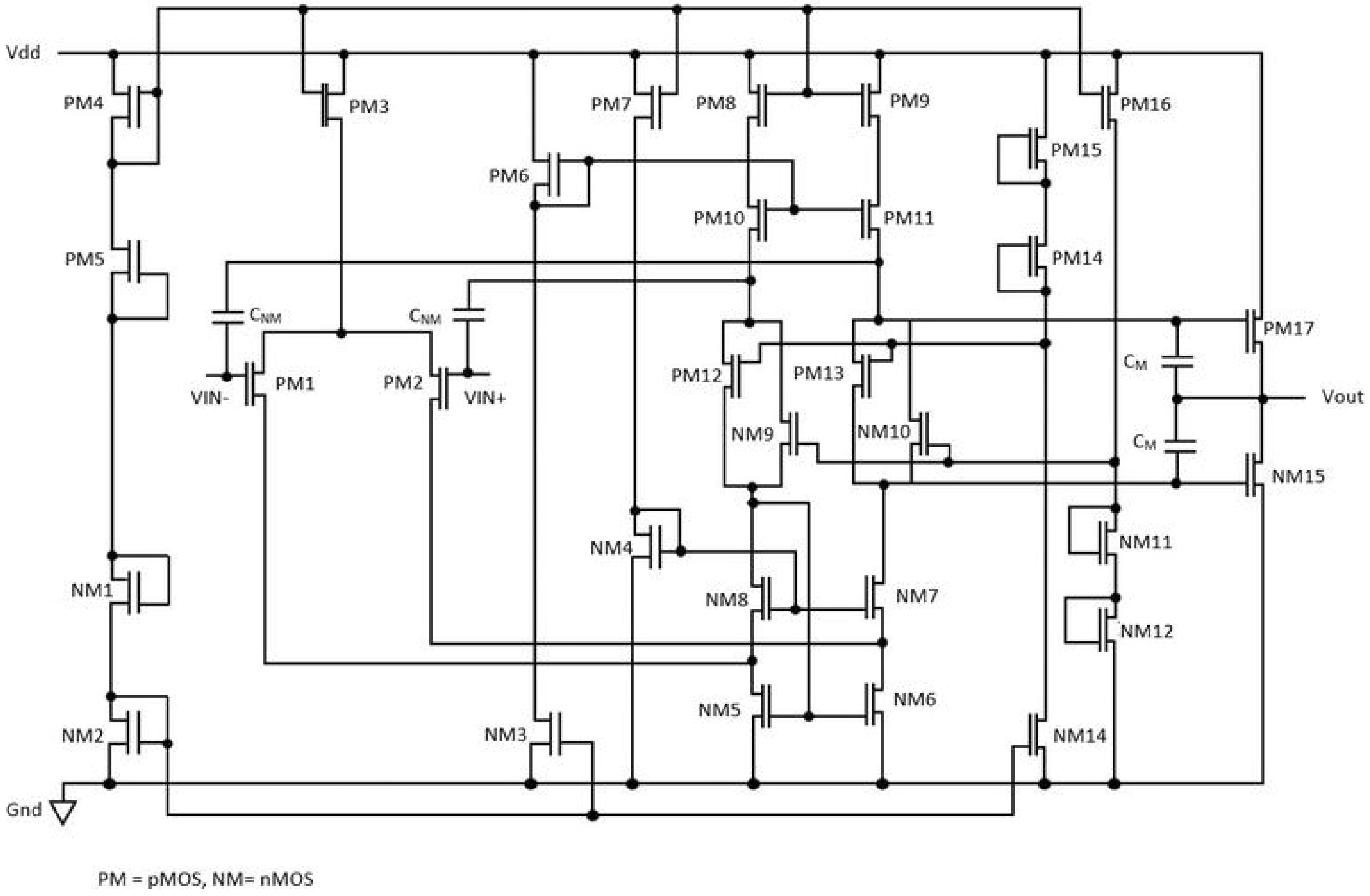

https://giselleyar.shop/product_details/28170740.html

- Need for a lot of (twenty-two) transistors and four capacitors

Operational Amplifiers: Non-Inverting Amplifier

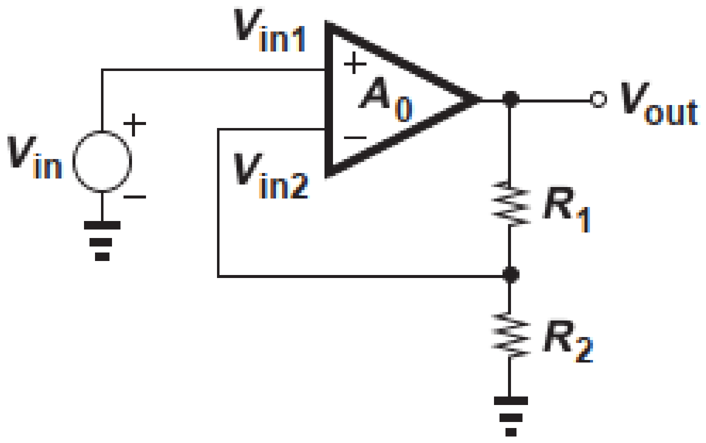

Figure 8.5 Noninverting amplifier.

$$V_{in2} = \frac{R_2}{R_1 + R_2} V_{out}$$

$$(V_{in1} - V_{in2}) A_0 = V_{out}$$

Razavi: “Fundamentals of Microelectronics”, 3rd edition, Wiley & Sons, 2021.

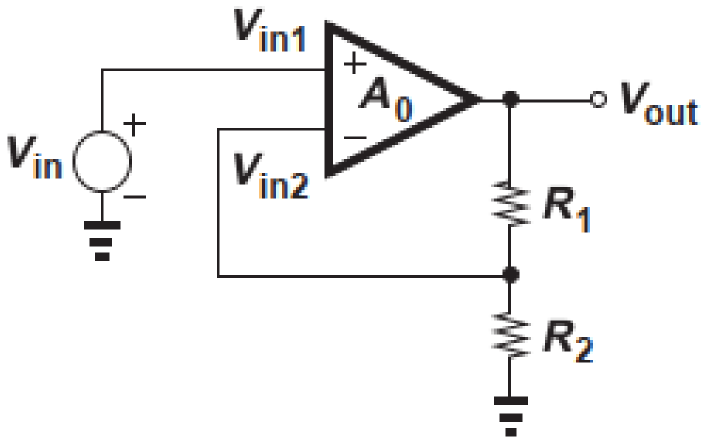

Figure 8.5 Noninverting amplifier.

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

$$V_{in2} = \frac{R_2}{R_1 + R_2} V_{out}$$

- For a relatively high gain:

$$V_{in1} \approx V_{in2}$$
$$\approx \frac{R_2}{R_1 + R_2} V_{out}$$

$$(V_{in1} - V_{in2})A_0 = V_{out}$$

$$v_{Id} = \frac{v_O}{A} = 0 \qquad \text{for } A = \infty$$

$$\frac{V_{out}}{V_{in}} \approx 1 + \frac{R_1}{R_2}$$

Operational Amplifiers: Non-Inverting Amplifier

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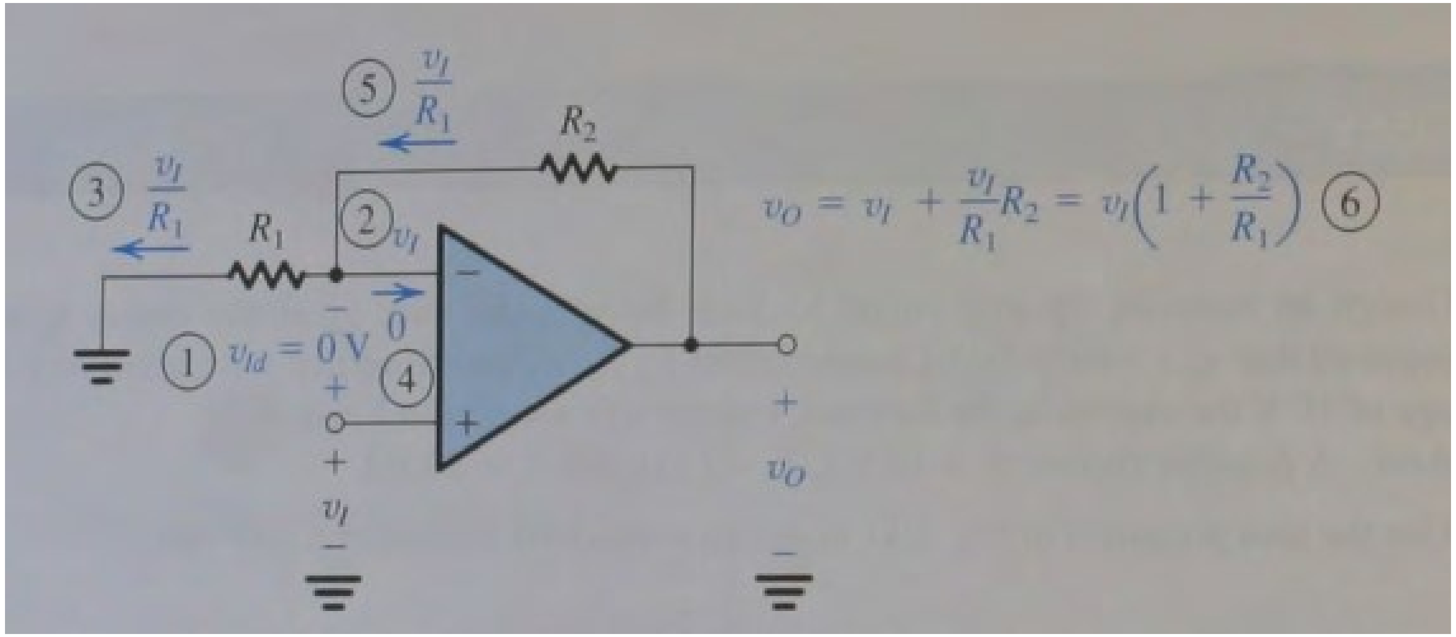

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

$$V_{in2} = \frac{R_2}{R_1 + R_2} V_{out}$$

- For an infinite gain:

$$(V_{in1} - V_{in2})A_0 = V_{out}$$

$$v_{Id} = \frac{v_O}{A} = 0 \qquad \text{for } A = \infty$$

$$v_O = v_I + \frac{v_I}{R_1} R_2 = v_I\left(1 + \frac{R_2}{R_1}\right)$$

Operational Amplifiers: Non-Inverting Amplifier

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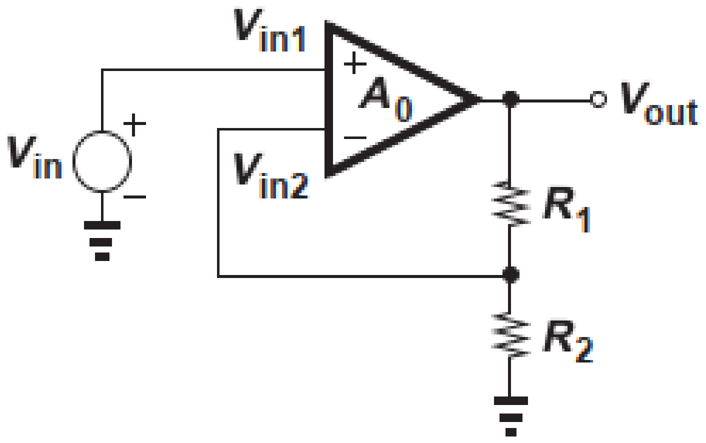

Figure 8.5 Noninverting amplifier.

Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

$$V_{in2} = \frac{R_2}{R_1 + R_2} V_{out}$$

$$(V_{in1} - V_{in2}) A_0 = V_{out}$$

$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + \frac{R_2}{R_1 + R_2} A_0}$$

$$\frac{V_{out}}{V_{in}} \approx \left(1 + \frac{R_1}{R_2}\right)\left[1 - \left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}\right]$$